2.7 Monthly Gridded Radiative Fluxes and Clouds (FSW)

EOSDIS Product Code: CER05

The Monthly Gridded Single Satellite Fluxes and Clouds (FSW) archival data product contains hourly single satellite flux and cloud parameters averaged over 1.0-degree regions. Input to the FSW Subsystem is the Clouds and Radiative Swath (CRS) archival data product. Each FSW covers a single month swath from a single CERES instrument mounted on one satellite. The product is written in HDF and contains metadata as well as gridded science data. The HDF product for TRMM consists of eight files, each containing data for ten 1.0-degree equal-angle zones. The HDF product for Terra and Aqua consists of 60 files, each containing data for three 1.0-degree equal-angle zones. Each record contains spatially averaged data for an individual region.

The major categories of data output on the FSW are as follows:

- Region data
- Imager Radiances Statistics
- Angular model scene classes
- Radiative fluxes for both Clear-sky and Total-sky at TOA
- Atmospheric flux profiles for Pristine-sky, Clear-sky and Total-sky
- Flux Adjustments for Pristine-sky, Clear-sky and Total-sky
- Surface Emissivity
- Overlap data for eleven cloud conditions
- Cloud category properties for four cloud layers
- Adjustment parameters for four cloud layers
- Adjustment parameters for clear-skies

A complete listing of parameters for this data product can be found in Tables 2.7-4 through Table 2.7-27.

Level: 3
Frequency: 1/Month
Portion of Atmosphere Covered: TOA, Surface, and Atmospheric Pressure Levels

Time Interval Covered:
File: Month
Record: Hour

Portion of Globe Covered:
File: Gridded Satellite Swath
Record: 1.0-Deg Equal-angle Regions

Product Version:
TRMM: Edition2C
Terra: Edition2A
Aqua: N/A

FSW Metadata

The types of FSW metadata are summarized in Table 2.7-1 and contain information which need only be recorded once per product. The CERES metadata are listed in Appendix B. The FSW product-specific metadata parameters are listed in Table 2.7-2.

Table 2.7-1. FSW Metadata Summary

HDF Name	Description Table	Records	Number of Fields
CERES Baseline Header Metadata	Table B-1	1	36
CERES_metadata Vdata	Table B-2	1	14
FSW Product Specific Metadata	Table 2.7-2	1	2

Table 2.7-2. FSW Product-specific Metadata

Item	Parameter Name	Description	DataType	Units	Range
1	ZoneBeginning	Beginning zone number	I4	N/A	1 .. 180
2	ZoneEnding	Ending zone number	I4	N/A	1 .. 180

All of the FSW science data are organized into the Vdata structures summarized in Table 2.7-3. The TRMM FSW product parameter lists are summarized in Table 2.7-4 through Table 2.7-13 and Table 2.7-18 through Table 2.7-24 including the SDS number, the SDS name, the data type, the units, the range, and the number of elements within each field. The Terra and Aqua FSW product parameter lists are summarized in Table 2.7-4 through Table 2.7-9 and Table 2.7-14 through Table 2.7-27 including the SDS number, the SDS name, the data type, the units, the range, and the number of elements within each field. Terra and Aqua FSW products contain additional MODIS aerosol SDSs which are summarized in Table 2.7-25 through Table 2.7-27. The profile flux SDS names are defined differently between TRMM and Terra FSW products, therefore the profile flux SDSs are summarized in separate tables for TRMM and Terra products. The number of records per Vdata is defined as n where n varies for each file. Sizing estimates are based on anticipated Terra sampling.

Table 2.7-3. FSW Vdata Summary (1 of 2)

Vdata Name	Description Table	Records	Number of Fields	VData Size (MB)
Time and Position Data	Table 2.7-4	n	6	311.04
Regional Identification Data	Table 2.7-5	n	5	115.20
Surface Map And Clear Area Data	Table 2.7-6	n	11	1555.20
Imager Radiances Statistics	Table 2.7-7	n	5	276.48
Angular Model Scene Type Data	Table 2.7-8	n	8	5235.84

Table 2.7-3. FSW Vdata Summary (2 of 2)

Vdata Name	Description Table	Records	Number of Fields	VData Size (MB)
TOA Fluxes	Table 2.7-9	n	12	1866.24
Pristine Vertical Flux Profiles - TRMM	Table 2.7-10	n	12	1866.24
Constrained Clear Sky Profiles - TRMM	Table 2.7-11	n	30	1555.20
Constrained Total Sky Profiles - TRMM	Table 2.7-12	n	30	1555.20
Constrainment - Initial Flux Deltas - TRMM	Table 2.7-13	n	27	1399.68
Pristine Vertical Flux Profiles - Terra	Table 2.7-14	n	12	1866.24
Constrained Clear Sky Profiles - Terra	Table 2.7-15	n	30	1555.20
Constrained Total Sky Profiles - Terra	Table 2.7-16	n	30	1555.20
Constrainment - Initial Flux Deltas - Terra	Table 2.7-17	n	27	1399.68
Surface Emissivity	Table 2.7-18	n	7	362.88
Cloud Overlap Conditions	Table 2.7-19	n	11	570.24
Cloud Layer - High	Table 2.7-20	n	18	2799.36
Cloud Layer - UpperMid	Table 2.7-21	n	18	2799.36
Cloud Layer - LowerMid	Table 2.7-22	n	18	2799.36
Cloud Layer - Low	Table 2.7-23	n	18	2799.36
Constrainment - Adjustments	Table 2.7-24	n	7	362.88
Aerosol LAND	Table 2.7-25	n	3	155.52
Aerosol OCEAN	Table 2.7-26	n	7	362.88
Cloudy Skies With No Aerosols	Table 2.7-27	n	21	1088.64
TOTAL SIZE				29856.80

Table 2.7-4. Time and Position Data

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
0	0	Julian Time	64-Bit Float[a]	day	2 440 000.0 .. 2 480 000.0	1	3
1	1	Sun Colatitude	32-Bit Float	deg	0.0 .. 180.0	1	3
2	2	Sun Longitude	32-Bit Float	deg	0.0 .. 360.0	1	3
3	3	Relative Azimuth Angle	32-Bit Float	deg	0.0 .. 360.0	1	3
4	4	Cos. Solar Zenith Angle	32-Bit Float	N/A	0.0 .. 1.0	1	3
5	5	Spacecraft Zenith Angle	32-Bit Float	deg	0.0 .. 90.0	1	3

a. Julian Time SDS Data Type is set to 32-Bit Float on TRMM products.

Table 2.7-5. Regional Identification Data

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
6	6	Region Number	32-Bit Float	N/A	1 .. 64800	1	3
7	7	Hour Box Number	32-Bit Float	N/A	1 .. 744	1	3
8	8	Num. Footprints in Region	32-Bit Float	N/A	1 .. 500	1	3
9	9	Colatitude	32-Bit Float	N/A	1 - 180	1	3
10	10	Longitude	32-Bit Float	N/A	1 - 360	1	3

Table 2.7-6. Surface Map And Clear Area Data

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
11	11	Alt. of Srf. above Sea	32-Bit Float	m	-1000 .. 10000	1	3
12	12	Surface Type Percentage	32-Bit Float	percent	0.0 .. 100.0	20	4
13	13	Snow/Ice Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
14	14	Smoke Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
15	15	Aerosol Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
16	16	Flag - Aerosol Type	32-Bit Float	N/A	0.0 .. 9999.0	1	3
17	17	Aerosol Opt. Depth at 0.63 µm in clr	32-Bit Float	µm	-1.0 .. 5.0	1	3
18	18	Aerosol Opt. Depth at 1.6 µm in clr	32-Bit Float	µm	-1.0 .. 5.0	1	3
19	19	Precipitable Water	32-Bit Float	cm	0.001 .. 10.0	1	3
20	20	Flag - Source Precip. H2O	32-Bit Float	N/A	0 .. 120	1	3
21	21	MOA - Relative Column Avg. Humidity	32-Bit Float	N/A	0.0 .. 100.0	1	3

Table 2.7-7. Imager Radiances Statistics (1 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
22	22	Imager Viewing Zenith Angle	32-Bit Float	deg	0.0 .. 90.0	1	3
23	23	Imager Relative Azimuth Angle	32-Bit Float	deg	0.0 .. 360.0	1	3
24	24	Imager Channel Central Wave-length	32-Bit Float	µm	0.4 .. 15.0	5	4

Table 2.7-7. Imager Radiances Statistics (2 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
25	25	Imager Radiance	32-Bit Float	$W\ m^{-2} sr^{-1} \mu m^{-1}$	-1000.0 .. 1000.0	5	4
26	26	Imager Radiance Clr-sky	32-Bit Float	$W\ m^{-2} sr^{-1} \mu m^{-1}$	-1000.0 .. 1000.0	5	4

Table 2.7-8. Angular Model Scene Type Data

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
27	27	Incident Solar Flux	32-Bit Float	N/A	0.0 .. 1400.0	1	3
28	N/A	Snow/Ice Percentage from Imager History	32-Bit Float	percent	0.0 .. 100.0	1	3
29	28	Area Coverage	32-Bit Float	percent	0.0 .. 100.0	20	4
30	29	SW Scene ID	32-Bit Float	N/A	0 .. 5000	20	4
31	30	Albedo (mean)	32-Bit Float	N/A	0.0 .. 1.0	20	4
32	31	Albedo (std)	32-Bit Float	N/A	0.0 .. 1.0	20	4
33	32	LW (mean)	32-Bit Float	$W\ m^{-2}$	0.0 .. 400.0	20	4
34	33	LW (std)	32-Bit Float	$W\ m^{-2}$	0.0 .. 400.0	20	4

Table 2.7-9. TOA Fluxes (1 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
35	34	SW TOA Clear-Sky	32-Bit Float	$W\ m^{-2}$	0.0 .. 1400.0	3	4
36	35	LW TOA Clear-Sky	32-Bit Float	$W\ m^{-2}$	0.0 .. 500.0	3	4
37	36	WN TOA Clear-Sky	32-Bit Float	$W\ m^{-2}$	0.0 .. 200.0	3	4
38	37	ALB TOA Clear-Sky	32-Bit Float	N/A	0.0 .. 1.0	3	4
39	38	SW TOA DOWNWARD Clear-Sky[a]	32-Bit Float	$W\ m^{-2}$	0.0 .. 1400.0	3	4
40	39	SW TOA UPWARD Clear-Sky[a]	32-Bit Float	$W\ m^{-2}$	0.0 .. 1400.0	3	4
41	40	LW TOA UPWARD Clear-Sky[a]	32-Bit Float	$W\ m^{-2}$	0.0 .. 850.0	3	4
42	41	WN TOA UPWARD Clear-Sky[a]	32-Bit Float	N/A	0.0 .. 370.0	3	4
43	42	SW TOA Total-Sky	32-Bit Float	$W\ m^{-2}$	0.0 .. 1400.0	3	4

Table 2.7-9. TOA Fluxes (2 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
44	43	LW TOA Total-Sky	32-Bit Float	W m^{-2}	0.0 .. 500.0	3	4
45	44	WN TOA Total-Sky	32-Bit Float	W m^{-2}	0.0 .. 200.0	3	4
46	45	ALB TOA Total-Sky	32-Bit Float	N/A	0.0 .. 1.0	3	4

a. SW, LW, WN upward & SW downward at TOA is reported in an additional set of SDSs for Clear Sky, so this could be directly comparable to CERES OBSERVED TOA Clear Sky Fluxes.

Table 2.7-10. Pristine Vertical Flux Profiles (TRMM)

Terra/Aqua SDS Index (See Table 2.7.14)	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	46	SW Upward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
	47	SW Downward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
	48	LW Upward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	0.0 - 850.0	3	4
	49	LW Downward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	0.0 - 700.0	3	4
	50	WN Upward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4
	51	WN Downward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4
	52	SW Upward Prestine Sky (SRF)	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
	53	SW Downward Prestine Sky (SRF)	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
	54	LW Upward Prestine Sky (SRF)	32-Bit Float	W m^{-2}	0.0 - 850.0	3	4
	55	LW Downward Prestine Sky (SRF)	32-Bit Float	W m^{-2}	0.0 - 700.0	3	4
	56	WN Upward Prestine Sky (SRF)	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4
	57	WN Downward Prestine Sky (SRF)	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4

Table 2.7-11. Constrained Clear Sky Profiles (1 of 2) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.15	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	58	SW Upward Clear Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	59	SW Downward Clear Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	60	LW Upward Clear Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	61	LW Downward Clear Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	62	WN Upward Clear Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	63	WN Downward Clear Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	64	SW Upward Clear Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	65	SW Downward Clear Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	66	LW Upward Clear Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	67	LW Downward Clear Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	68	WN Upward Clear Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	69	WN Downward Clear Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	70	SW Upward Clear Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	71	SW Downward Clear Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	72	LW Upward Clear Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	73	LW Downward Clear Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	74	WN Upward Clear Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	75	WN Downward Clear Sk (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	76	SW Upward Clear Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	77	SW Downward Clear Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4

Table 2.7-11. Constrained Clear Sky Profiles (2 of 2) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.15	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	78	LW Upward Clear Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	79	LW Downward Clear Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	80	WN Upward Clear Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	81	WN Downward Clear Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	82	SW Upward Clear Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	83	SW Downward Clear Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	84	LW Upward Clear Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	85	LW Downward Clear Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	86	WN Upward Clear Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	87	WN Downward Clear Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4

Table 2.7-12. Constrained Total Sky Profiles (1 of 3) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.16)	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	88	SW Upward Total Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	89	SW Downward Total Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	90	LW Upward Total Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	91	LW Downward Total Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	92	WN Upward Total Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	93	WN Downward Total Sky (TOA)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	94	SW Upward Total Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4

Table 2.7-12. Constrained Total Sky Profiles (2 of 3) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.16)	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	95	SW Downward Total Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	96	LW Upward Total Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	97	LW Downward Total Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	98	WN Upward Total Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	99	WN Downward Total Sky (70hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	100	SW Upward Total Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	101	SW Downward Total Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	102	LW Upward Total Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	103	LW Downward Total Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	104	WN Upward Total Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	105	WN Downward Total Sky (200hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	106	SW Upward Total Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	107	SW Downward Total Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	108	LW Upward Total Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	109	LW Downward Total Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
	110	WN Upward Total Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	111	WN Downward Total Sky (500hPa)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	112	SW Upward Total Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	113	SW Downward Total Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
	114	LW Upward Total Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
	115	LW Downward Total Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4

Table 2.7-12. Constrained Total Sky Profiles (3 of 3) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.16)	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	116	WN Upward Total Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
	117	WN Downward Total Sky (SFC)	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4

Table 2.7-13. Constrainment - Initial Flux Deltas (1 of 2) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.17)	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	118	SW Upward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	119	LW Upward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
	120	WN Upward Prestine Sky (TOA)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	121	SW Upward Prestine Sky (SFC)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	122	SW Downward Prestine Sky (SFC)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	123	LW Upward Prestine Sky (SFC)	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
	124	LW Downward Prestine Sky (SFC)	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
	125	WN Upward Prestine Sky (SFC)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	126	WN Downward Prestine Sky (SFC)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	127	SW Upward Clear Sky (TOA)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	128	LW Upward Clear Sky (TOA)	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
	129	WN Upward Clear Sky (TOA)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	130	SW Upward Clear Sky (SFC)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	131	SW Downward Clear Sky (SFC)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	132	LW Upward Clear Sky (SFC)	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4

Table 2.7-13. Constrainment - Initial Flux Deltas (2 of 2) (TRMM)

Terra/Aqua SDS Index (See Table 2.7.17)	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
	133	LW Downward Clear Sky (SFC)	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
	134	WN Upward Clear Sky (SFC)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	135	WN Downward Clear Sky (SFC)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	136	SW Upward Total Sky (TOA)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	137	LW Upward Total Sky (TOA)	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
	138	WN Upward Total Sky (TOA)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	139	SW Upward Total Sky (SFC)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	140	SW Downward Total Sky (SFC)	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
	141	LW Upward Total-Sky (SFC)	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
	142	LW Downward Total-Sky (SFC)	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
	143	WN Upward Total-Sky (SFC)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
	144	WN Downward Total-Sky (SFC)	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4

Table 2.7-14. Pristine Vertical Flux Profiles (1 of 2) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.10)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
47		SW flux - upward - pristine - TOA	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
48		SW flux - upward - pristine - surfa	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
49		SW flux - downward - pristine - TOA	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
50		SW flux - downward - pristine - sur	32-Bit Float	W m^{-2}	0.0 - 1400.0	3	4
51		LW flux - upward - pristine - TOA	32-Bit Float	W m^{-2}	0.0 - 850.0	3	4
52		LW flux - upward - pristine - surfa	32-Bit Float	W m^{-2}	0.0 - 850.0	3	4

Table 2.7-14. Pristine Vertical Flux Profiles (2 of 2) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.10)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
53		LW flux - downward - pristine - TOA	32-Bit Float	W m^{-2}	0.0 - 700.0	3	4
54		LW flux - downward - pristine - sur	32-Bit Float	W m^{-2}	0.0 - 700.0	3	4
55		WN flux - upward - pristine - TOA	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4
56		WN flux - upward - pristine - surfa	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4
57		WN flux - downward - pristine - TOA	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4
58		WN flux - downward - pristine - sur	32-Bit Float	W m^{-2}	0.0 - 370.0	3	4

Table 2.7-15. Constrained Clear Sky Profiles (1 of 2) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.11)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
59		SW flux - upward - clear-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
60		SW flux - upward - clear-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
61		SW flux - upward - clear-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
62		SW flux - upward - clear-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
63		SW flux - upward - clear-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
64		SW flux - downward - clear-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
65		SW flux - downward - clear-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
66		SW flux - downward - clear-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
67		SW flux - downward - clear-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
68		SW flux - downward - clear-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4

Table 2.7-15. Constrained Clear Sky Profiles (2 of 2) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.11)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
69		LW flux - upward - clear-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
70		LW flux - upward - clear-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
71		LW flux - upward - clear-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
72		LW flux - upward - clear-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
73		LW flux - upward - clear-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
74		LW flux - downward - clear-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
75		LW flux - downward - clear-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
76		LW flux - downward - clear-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
77		LW flux - downward - clear-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
78		LW flux - downward - clear-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
79		WN flux - upward - clear-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
80		WN flux - upward - clear-sky - 70hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
81		WN flux - upward - clear-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
82		WN flux - upward - clear-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
83		WN flux - upward - clear-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
84		WN flux - downward - clear-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
85		WN flux - downward - clear-sky - 70hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
86		WN flux - downward - clear-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
87		WN flux - downward - clear-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
88		WN flux - upward - clear-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4

Table 2.7-16. Constrained Total Sky Profiles (1 of 2) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.12)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
89		SW flux - upward - total-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
90		SW flux - upward - total-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
91		SW flux - upward - total-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
92		SW flux - upward - total-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
93		SW flux - upward - total-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
94		SW flux - downward - total-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
95		SW flux - downward - total-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
96		SW flux - downward - total-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
97		SW flux - downward - total-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
98		SW flux - downward - total-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
99		LW flux - upward - total-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
100		LW flux - upward - total-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
101		LW flux - upward - total-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
102		LW flux - upward - total-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
103		LW flux - upward - total-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
104		LW flux - downward - total-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
105		LW flux - downward - total-sky - 70hPA	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
106		LW flux - downward - total-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
107		LW flux - downward - total-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4

Table 2.7-16. Constrained Total Sky Profiles (2 of 2) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.12)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
108		LW flux - downward - total-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
109		WN flux - upward - total-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
110		WN flux - upward - total-sky - 70hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
111		WN flux - upward - total-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
112		WN flux - upward - total-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
113		WN flux - upward - total-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
114		WN flux - downward - total-sky - TOA	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
115		WN flux - downward - total-sky - 70hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
116		WN flux - downward - total-sky - 200hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
117		WN flux - downward - total-sky - 500hPa	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
118		WN flux - upward - total-sky - SFC	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4

Table 2.7-17. Constrainment - Initial Flux Deltas (1 of 3) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.13)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
119		SW flux adjustment at surface upward - pristine	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
120		SW flux adjustment at TOA - upward - pristine	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
121		SW flux adjustment at surface - downward - pristine	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
122		LW flux adjustment at surface - upward - pristine	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4

Table 2.7-17. Constrainment - Initial Flux Deltas (2 of 3) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.13)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
123		LW flux adjustment at surface - downward - pristine	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
124		LW flux adjustment at TOA - upward - pristine	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
125		WN flux adjustment at surface - upward - pristine	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
126		WN flux adjustment at surface - downward - pristine	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
127		WN flux adjustment at TOA - upward - pristine	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
128		SW flux adjustment at surface upward - clear-sky	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
129		SW flux adjustment at TOA - upward - clear-sky	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
130		SW flux adjustment at surface - downward - clear-sky	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
131		LW flux adjustment at surface - upward - clear-sky	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
132		LW flux adjustment at surface - downward - clear-sky	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
133		LW flux adjustment at TOA - upward - clear-sky	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
134		WN flux adjustment at surface - upward - clear-sky	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
135		WN flux adjustment at surface - downward - clear-sky	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
136		WN flux adjustment at TOA - upward - clear-sky	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
137		SW flux adjustment at surface upward - total-sky	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
138		SW flux adjustment at TOA - upward - total-sky	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
139		SW flux adjustment at surface - downward - total-sky	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
140		LW flux adjustment at surface - upward - total-sky	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
141		LW flux adjustment at surface - downward - total-sky	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
142		LW flux adjustment at TOA - upward - total-sky	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4

Table 2.7-17. Constrainment - Initial Flux Deltas (3 of 3) (Terra)

Terra/ Aqua SDS Index	TRMM SDS Index (See Table 2.7.13)	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
143		WN flux adjustment at surface - upward - total-sky	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
144		WN flux adjustment at surface - downward - total-sky	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
145		WN flux adjustment at TOA - upward - total-sky	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4

Table 2.7-18. Surface Emissivity

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
146	145	LW Surface Emissivity	32-Bit Float	N/A	0.0 .. 1.0	1	3
147	146	WN Surface Emissivity	32-Bit Float	N/A	0.0 .. 1.0	1	3
148	147	Photo. Syn. Radiation	32-Bit Float	W m^{-2}	0.0 .. 780.0	1	3
149	148	Direct/Diffuse	32-Bit Float	N/A	0.0 .. 30.0	1	3
150	149	Initial Broadband Albedo	32-Bit Float	N/A	0.0 .. 1.0	1	3
151	150	Surface Albedo	32-Bit Float	N/A	0.0 .. 1.0	1	3
152	151	Skin Temperature	32-Bit Float	K	175.0 .. 375.0	1	3

Table 2.7-19. Cloud Overlap Conditions (1 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
153	152	Clear	32-Bit Float	percent	0.0 .. 100.0	1	3
154	153	Low	32-Bit Float	percent	0.0 .. 100.0	1	3
155	154	Lowermid	32-Bit Float	percent	0.0 .. 100.0	1	3
156	155	Uppermid	32-Bit Float	percent	0.0 .. 100.0	1	3
157	156	High	32-Bit Float	percent	0.0 .. 100.0	1	3
158	157	High Uppermid	32-Bit Float	percent	0.0 .. 100.0	1	3
159	158	High Lowermid	32-Bit Float	percent	0.0 .. 100.0	1	3
160	159	High Low	32-Bit Float	percent	0.0 .. 100.0	1	3
161	160	Uppermid - Lowermid	32-Bit Float	percent	0.0 .. 100.0	1	3

Table 2.7-19. Cloud Overlap Conditions (2 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
162	161	Uppermid - Low	32-Bit Float	percent	0.0 .. 100.0	1	3
163	162	Lowermid - Low	32-Bit Float	percent	0.0 .. 100.0	1	3

Table 2.7-20. Cloud Layer - High (mean std num_obs)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
164	163	Area Fraction Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
165	164	Effective Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
166	165	Effective Temperature	32-Bit Float	K	100.0 .. 350.0	3	4
167	166	Effective Height	32-Bit Float	km	0.0 .. 20.0	3	4
168	167	Top Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
169	168	Bottom Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
170	169	Particle Phase	32-Bit Float	N/A	1.0 .. 2.0	3	4
171	170	Liquid Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
172	171	Ice Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
173	172	Liquid Particle Radius	32-Bit Float	µm	0.0 .. 40.0	3	4
174	173	Ice Particle Diameter	32-Bit Float	µm	0.0 .. 300.0	3	4
175	174	Vis. Opt. Depth (linear)	32-Bit Float	N/A	0.0 .. 400.0	3	4
176	175	Vis. Opt. Depth (log)	32-Bit Float	N/A	-6.0 .. 6.0	3	4
177	176	Infrared Emissivity	32-Bit Float	N/A	0.0 .. 2.0	3	4
178	177	Vertical Aspect Ratio	32-Bit Float	N/A	0.0 .. 20.0	3	4
179	178	Adj. Vis. Opt. Depth	32-Bit Float	N/A	-400.0 .. 400.0	1	3
180	179	Adj. Fractional Area	32-Bit Float	N/A	-1 ..1	1	3
181	180	Adj. Effective Temp.	32-Bit Float	K	TBD	1	3

Table 2.7-21. Cloud Layer - UpperMid (mean std num_obs) (1 of 2)

TRMM SDS Index	Terra/ Aqua SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
182	181	Area Fraction Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
183	182	Effective Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
184	183	Effective Temperature	32-Bit Float	K	100.0 .. 350.0	3	4
185	184	Effective Height	32-Bit Float	km	0.0 .. 20.0	3	4

Table 2.7-21. Cloud Layer - UpperMid (mean std num_obs) (2 of 2)

TRMM SDS Index	Terra/ Aqua SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
186	185	Top Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
187	186	Bottom Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
188	187	Particle Phase	32-Bit Float	N/A	1.0 .. 2.0	3	4
189	188	Liquid Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
190	189	Ice Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
191	190	Liquid Particle Radius	32-Bit Float	µm	0.0 .. 40.0	3	4
192	191	Ice Particle Diameter	32-Bit Float	µm	0.0 .. 300.0	3	4
193	192	Vis. Opt. Depth (linear)	32-Bit Float	N/A	0.0 .. 400.0	3	4
194	193	Vis. Opt. Depth (log)	32-Bit Float	N/A	-6.0 .. 6.0	3	4
195	194	Infrared Emissivity	32-Bit Float	N/A	0.0 .. 2.0	3	4
196	195	Vertical Aspect Ratio	32-Bit Float	N/A	0.0 .. 20.0	3	4
197	196	Adj. Vis. Opt. Depth	32-Bit Float	N/A	-400.0 .. 400.0	1	3
198	197	Adj. Fractional Area	32-Bit Float	N/A	-1 ..1	1	3
199	198	Adj. Effective Temp.	32-Bit Float	K	TBD	1	3

Table 2.7-22. Cloud Layer - LowerMid (mean std num_obs) (1 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
200	199	Area Fraction Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
201	200	Effective Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
202	201	Effective Temperature	32-Bit Float	K	100.0 .. 350.0	3	4
203	202	Effective Height	32-Bit Float	km	0.0 .. 20.0	3	4
204	203	Top Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
205	204	Bottom Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
206	205	Particle Phase	32-Bit Float	N/A	1.0 .. 2.0	3	4
207	206	Liquid Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
208	207	Ice Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
209	208	Liquid Particle Radius	32-Bit Float	µm	0.0 .. 40.0	3	4
210	209	Ice Particle Diameter	32-Bit Float	µm	0.0 .. 300.0	3	4
211	210	Vis. Opt. Depth (linear)	32-Bit Float	N/A	0.0 .. 400.0	3	4
212	211	Vis. Opt. Depth (log)	32-Bit Float	N/A	-6.0 .. 6.0	3	4
213	212	Infrared Emissivity	32-Bit Float	N/A	0.0 .. 2.0	3	4
214	213	Vertical Aspect Ratio	32-Bit Float	N/A	0.0 .. 20.0	3	4

Table 2.7-22. Cloud Layer - LowerMid (mean std num_obs) (2 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
215	214	Adj. Vis. Opt. Depth	32-Bit Float	N/A	-400.0 .. 400.0	1	3
216	215	Adj. Fractional Area	32-Bit Float	N/A	-1 .. 1	1	3
217	216	Adj. Effective Temp.	32-Bit Float	K	TBD	1	3

Table 2.7-23. Cloud Layer - Low (mean std num_obs)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
218	217	Area Fraction Percentage	32-Bit Float	percent	0.0 .. 100.0	1	3
219	218	Effective Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
220	219	Effective Temperature	32-Bit Float	K	100.0 .. 350.0	3	4
221	220	Effective Height	32-Bit Float	km	0.0 .. 20.0	3	4
222	221	Top Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
223	222	Bottom Pressure	32-Bit Float	hPa	0.0 .. 1100.0	3	4
224	223	Particle Phase	32-Bit Float	N/A	1.0 .. 2.0	3	4
225	224	Liquid Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
226	225	Ice Water Path	32-Bit Float	gm^{-2}	0.0 .. 10000.0	3	4
227	226	Liquid Particle Radius	32-Bit Float	µm	0.0 .. 40.0	3	4
228	227	Ice Particle Diameter	32-Bit Float	µm	0.0 .. 300.0	3	4
229	228	Vis. Opt. Depth (linear)	32-Bit Float	N/A	0.0 .. 400.0	3	4
230	229	Vis. Opt. Depth (log)	32-Bit Float	N/A	-6.0 .. 6.0	3	4
231	230	Infrared Emissivity	32-Bit Float	N/A	0.0 .. 2.0	3	4
232	231	Vertical Aspect Ratio	32-Bit Float	N/A	0.0 .. 20.0	3	4
233	232	Adj. Vis. Opt. Depth	32-Bit Float	N/A	-400.0 .. 400.0	1	3
234	233	Adj. Fractional Area	32-Bit Float	N/A	-1 .. 1	1	3
235	234	Adj. Effective Temp.	32-Bit Float	K	TBD	1	3

Table 2.7-24. Constrainment - Adjustments (mean) (1 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
236	235	Init. Precipitable H2O	32-Bit Float	cm	0 .. 10	1	3

Table 2.7-24. Constrainment - Adjustments (mean) (2 of 2)

Terra/ Aqua SDS Index	TRMM SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
237	236	Adj. Precipitable H2O	32-Bit Float	cm	-10 .. 10	1	3
238	237	Adj. Surface Albedo	32-Bit Float	N/A	-1 .. 1	1	3
239	238	Init. Aerosol Opt. Dep.	32-Bit Float	N/A	0 .. 2	1	3
240	239	Adj. Aerosol Opt. Dep.	32-Bit Float	N/A	-2 .. 2	1	3
241	240	Init. Skin Temperature	32-Bit Float	K	TBD	1	3
242	241	Adj. Skin Temperature	32-Bit Float	K	TBD	1	3

Table 2.7-25. Aerosol LAND

Terra/ Aqua SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
243	cor_optdepth047_land	32-Bit Float	N/A	0.0..5.0	1	3
244	cor_optdepth055_land	32-Bit Float	N/A	0.0..5.0	1	3
245	cor_optdepth066_land	32-Bit Float	N/A	0.0..5.0	1	3

Table 2.7-26. Aerosol OCEAN

Terra/ Aqua SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
246	eff_optdepth047_ocean	32-Bit Float	N/A	0.0..5.0	1	3
247	eff_optdepth055_ocean	32-Bit Float	N/A	0.0..5.0	1	3
248	eff_optdepth066_ocean	32-Bit Float	N/A	0.0..5.0	1	3
249	eff_optdepth087_ocean	32-Bit Float	N/A	0.0..5.0	1	3
250	eff_optdepth124_ocean	32-Bit Float	N/A	0.0..5.0	1	3
251	eff_optdepth164_ocean	32-Bit Float	N/A	0.0..5.0	1	3
252	eff_optdepth213_ocean	32-Bit Float	N/A	0.0..5.0	1	3

Table 2.7-27. Cloudy Skies with No Aerosols (1 of 2)

Terra/ Aqua SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
253	SW flux - upward cloudy skies with no aerosol - TOA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
254	SW flux - upward cloudy skies with no aerosol - surface	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
255	SW flux - downward cloudy skies with no aerosol - TOA	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
256	SW flux - downward cloudy skies with no aerosol - surface	32-Bit Float	W m^{-2}	0.0 .. 1400.0	3	4
257	LW flux - upward cloudy skies with no aerosol - TOA	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
258	LW flux - upward cloudy skies with no aerosol - surface	32-Bit Float	W m^{-2}	0.0 .. 850.0	3	4
259	LW flux - downward cloudy skies with no aerosol - TOA	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
260	LW flux - downward cloudy skies with no aerosol - surface	32-Bit Float	W m^{-2}	0.0 .. 700.0	3	4
261	WN flux - upward cloudy skies with no aerosol - TOA	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
262	WN flux - upward cloudy skies with no aerosol - surface	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
263	WN flux - downward cloudy skies with no aerosol - TOA	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
264	WN flux - downward cloudy skies with no aerosol - surface	32-Bit Float	W m^{-2}	0.0 .. 370.0	3	4
265	SW flux adjustments at surface-upward-cloudy skies/no aerosol	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
266	SW flux adjustments at TOA-upward-cloudy skies/no aerosol	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
267	SW flux adjustments at surface-downward-cloudy skies/no aero-sol	32-Bit Float	W m^{-2}	-1400.0 .. 1400.0	3	4
268	LW flux adjustments at surface-upward-cloudy skies/no aerosol	32-Bit Float	W m^{-2}	-600.0 .. 600.0	3	4
269	LW flux adjustments at surface-downward-cloudy skies/no aero-sol	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
270	LW flux adjustments at TOA-upward-cloudy skies/no aerosol	32-Bit Float	W m^{-2}	-700.0 .. 700.0	3	4
271	WN flux adjustments at surface-upward-cloudy skies/no aerosol	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4

Table 2.7-27. Cloudy Skies with No Aerosols (2 of 2)

Terra/ Aqua SDS Index	SDS Name / Parameter	Data Type	Units	Range	No. of Ele-ments	SDS rank/ dim
272	WN flux adjustments at surface-downward- cloudy skies/no aero-sol	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4
273	WN flux adjustments at TOA-upward-cloudy skies/ no aerosol	32-Bit Float	W m^{-2}	-50.0 .. 50.0	3	4

FSW Revision Record

The product Revision Record contains information pertaining to approved section changes. The table lists the date the Software Configuration Change Request (SCCR) was approved, the Release and Version Number, the SCCR number, a short description of the revision, and the revised sections. The authors are listed on the document cover.

FSW Revision Record

SCCR Approval Date	Release/ Version Number	SCCR Number	Description of Revision	Section(s) Affected
N/A	R3V1	N/A	• Updated format to comply with standards.	All
02/07/03	R3V2	424	• Updated parameters in Regional Identification Data, Regional Imager Data. • Updated Angular model scene category • Updated vdata sizes. • Added MODIS aerosols for land and Ocean. • Updated format to comply with standards.	Tables 2.11-5, 2.11-7 Table 2.7-8 Table 2.7-3 Tables 2.7-21, 2.7-22 All
10/23/03	R3V3	476	• Added cloudy skies with no Aerosols. • Updated flux profile parameters. • Added Snow/Ice Percentage from Imager History parameter to Angular Model Scene type. • Updated format to comply with standards.	Tables 2.7.3, 2.7-23 Tables 2.7-11 - 2.7-13 Table 2.7.8 All
10/23/03 01/28/04	R3V4	476 499	• Added two new columns to include SDS index numbers for TRMM & Terra. • Julian Time SDS data type is set to 64-bit. Added a note. • Added new tables for TRMM profile flux SDSs, since these names are defined differently for TRMM and Terra. • Updated units and range values. • Updated format to comply with standards.	All Tables 2.7.4 Tables 2.7.10 - 2.7.13 Tables 2.7.19 - 2.7.23 & 2.7.25-2.7.26 All